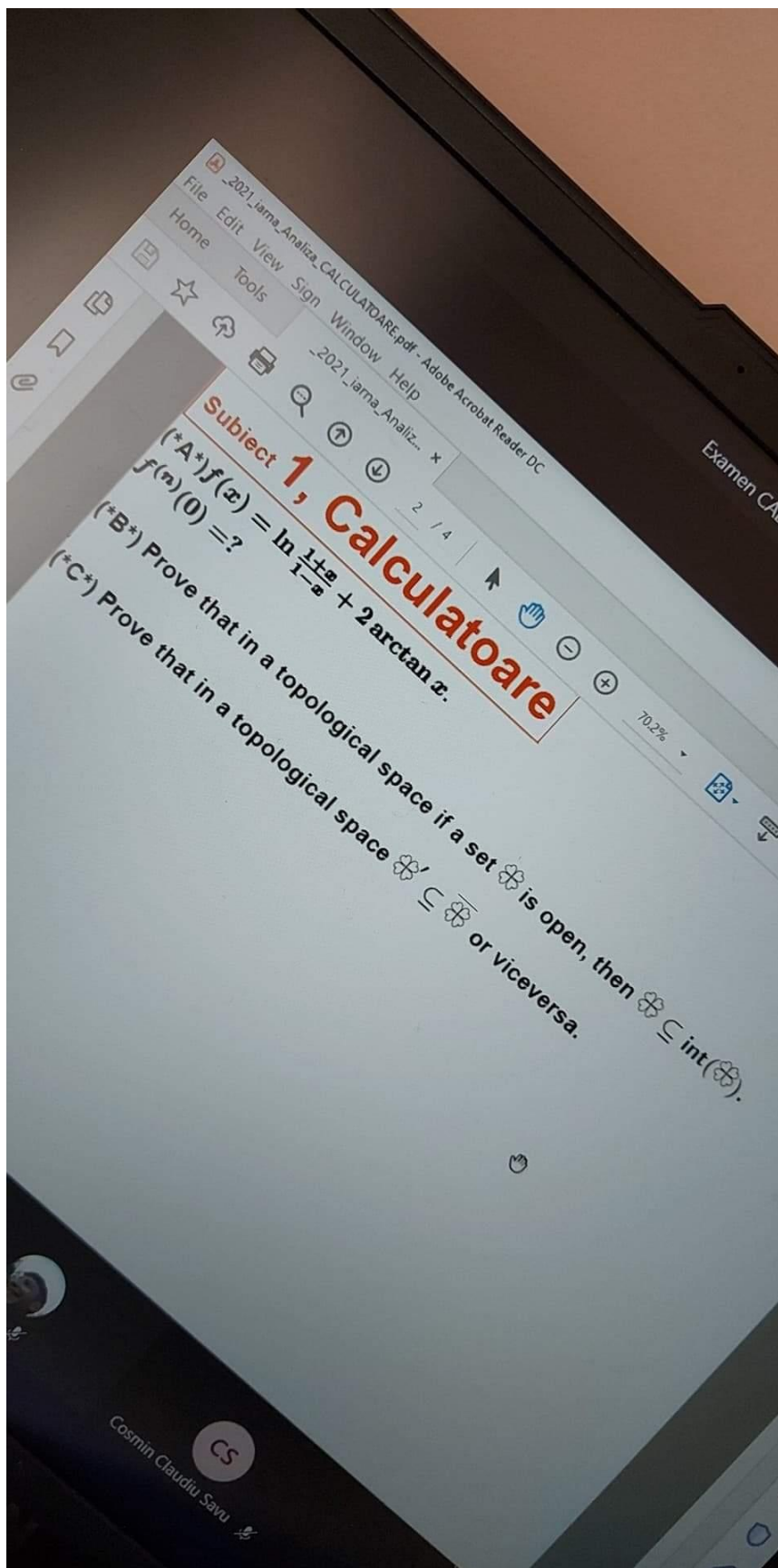
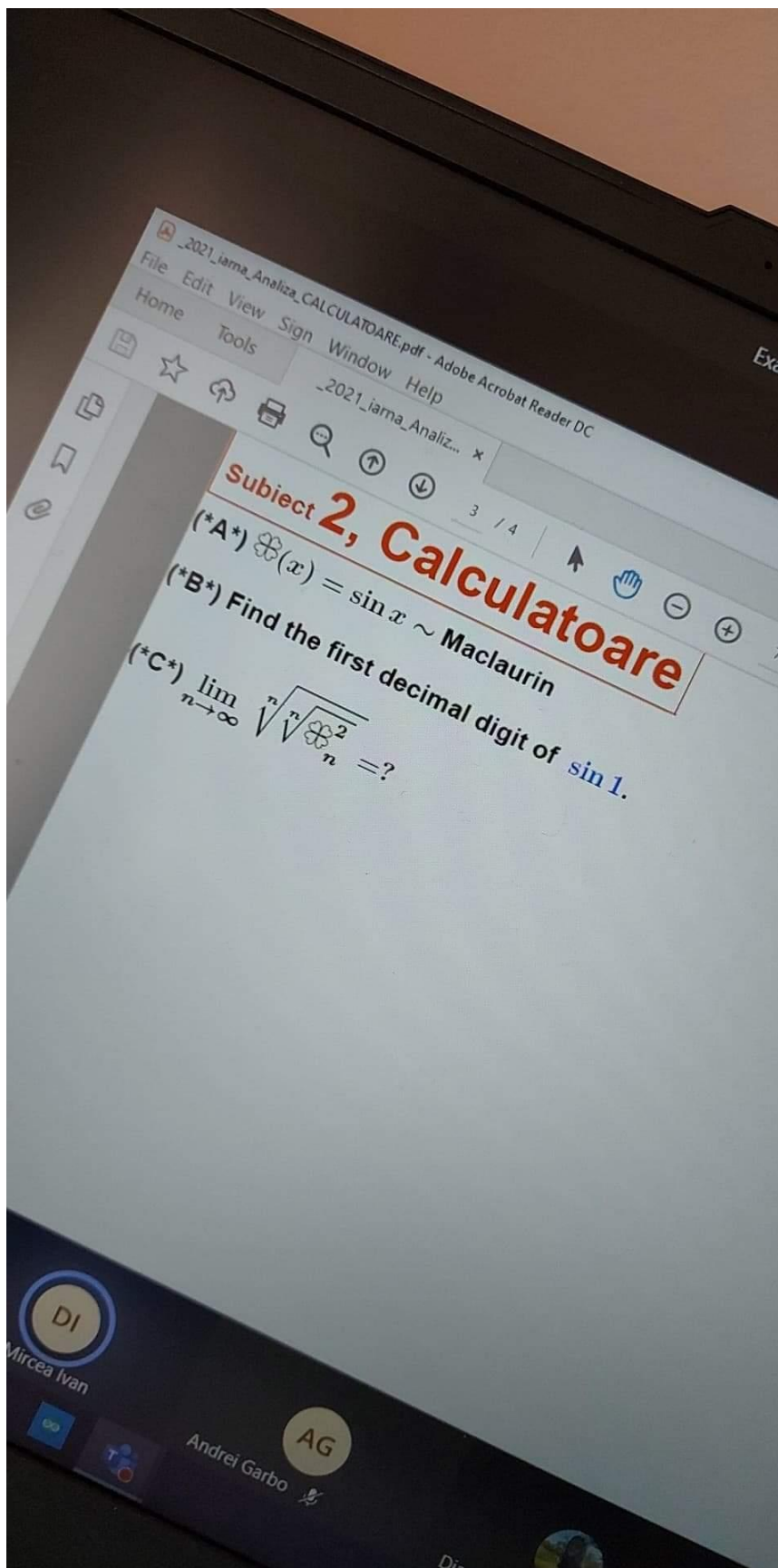
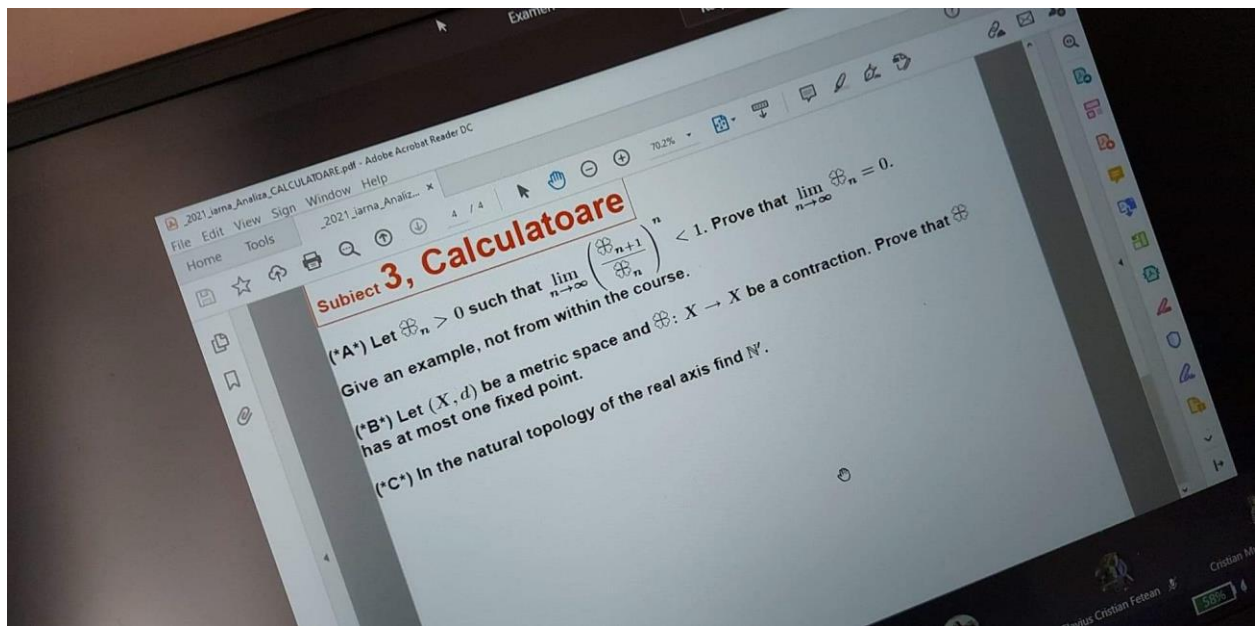
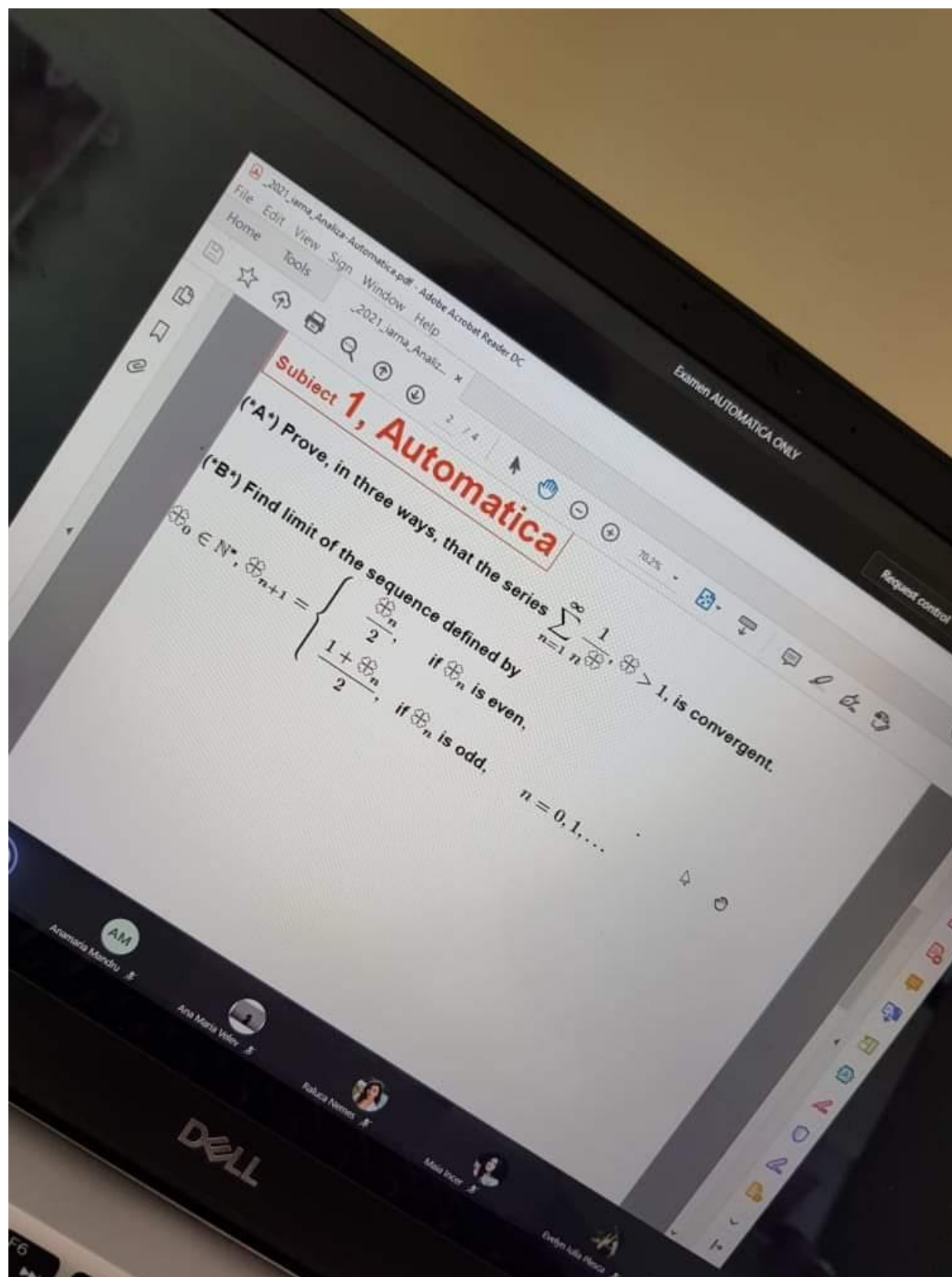
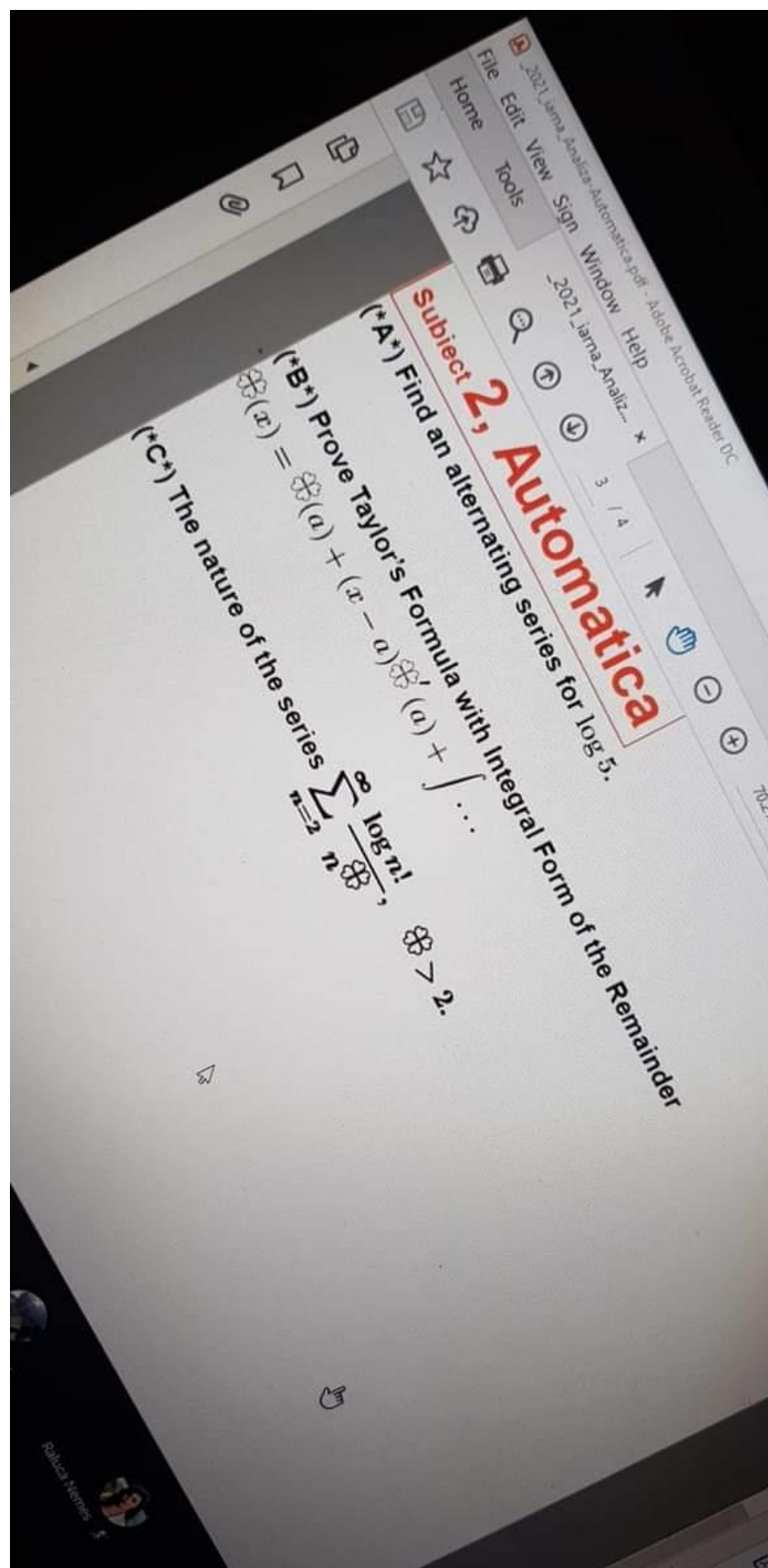












Subject 3, Automatica

(*A*) Let $\oplus_i(x) = \sum_{k=1}^n a_{i,k} x^k$, $i = 0, \dots, n$, $x \in \mathbb{R}$. Calculate the determinant

$$\begin{vmatrix} \oplus_0(x) & \oplus_1(x) & \dots & \oplus_n(x) \\ \oplus'_0(x) & \oplus'_1(x) & \dots & \oplus'_n(x) \\ \vdots & \vdots & \ddots & \vdots \\ \oplus^{(n)}_0(x) & \oplus^{(n)}_1(x) & \dots & \oplus^{(n)}_n(x) \end{vmatrix}, \quad x \in \mathbb{R}. \quad \checkmark$$

(*B*) Find the equation of the tangent plane at the point a of the surface given by $\oplus(x) = 0$.

(*C*) Find the first decimal digit of $\int_0^1 e^{-x^4} dx$.